# Trend research — 2026-04-19 · incr 20

always_in two-axis sweep — W=10, K=2 strictly dominates production W=5, K=2

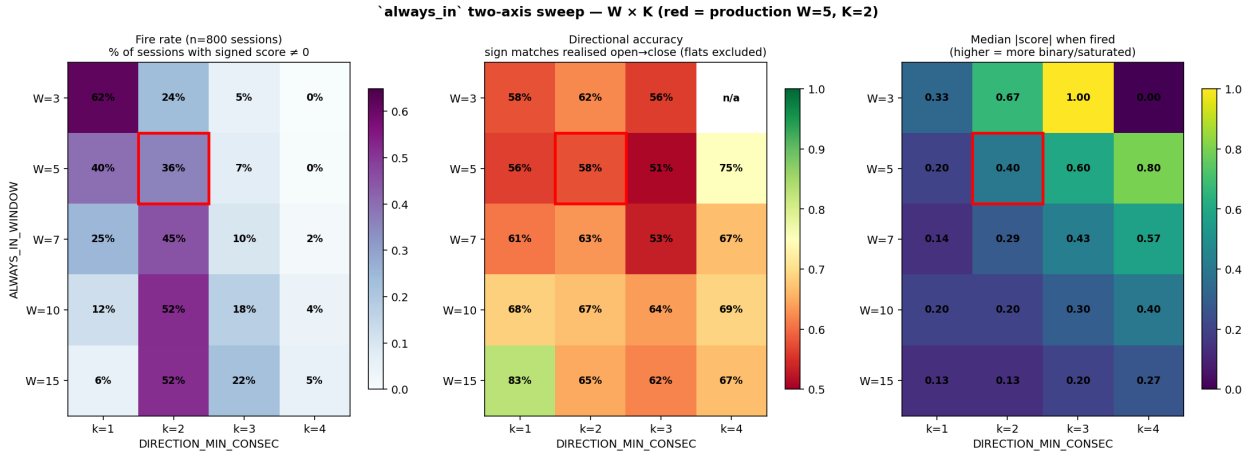
## TL;DR

Incr 19 flagged **always_in** as the weakest informative contributor — **36.0%** fire rate on 800 real RTH sessions at only **57.6%** directional accuracy, *below* the **66.0%** always-up baseline. Swept the two knobs (**ALWAYS_IN_WINDOW** × **DIRECTION_MIN_CONSEC**) over 20 cells on the same sample. Result: **W=10, K=2** strictly dominates production — fire rate **51.9%** (+16 pp), directional accuracy **66.9%** (+9 pp). Both above baseline. Read-only — recommendation needs Will's nod.

## Headline numbers

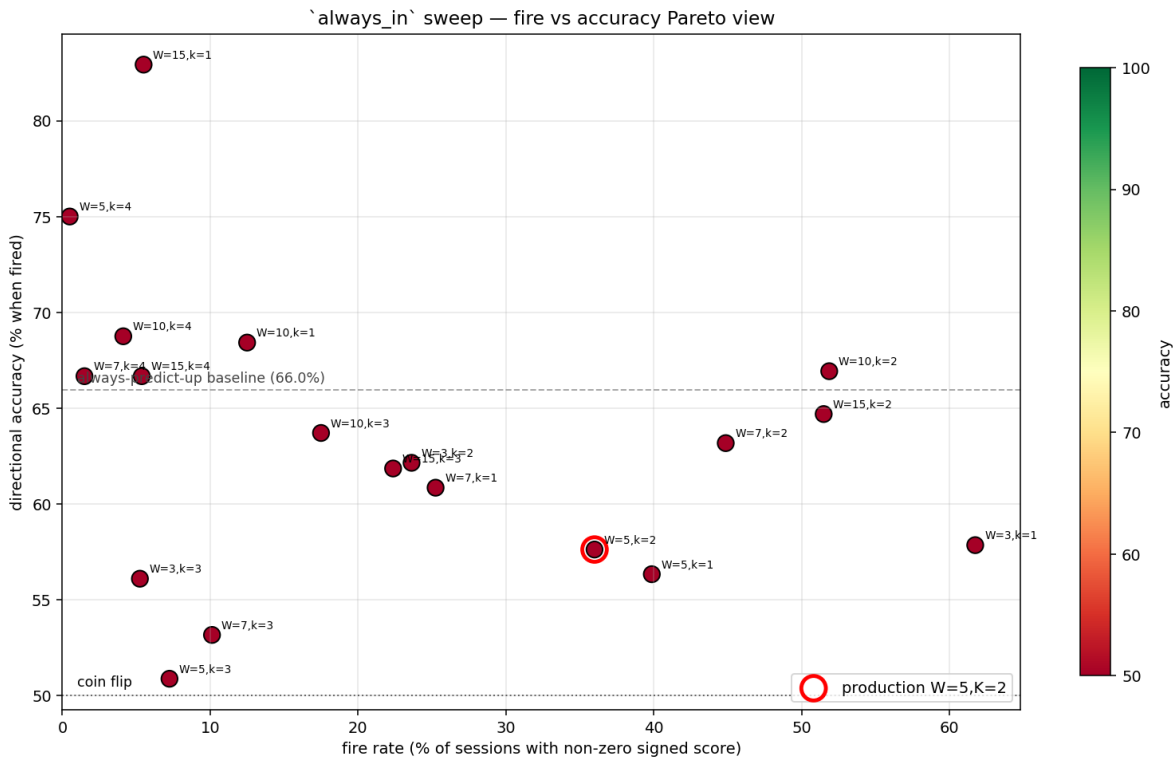
- Sample: **800 (symbol, session) pairs**, 387 unique symbols. Realised: up=494, down=255, flat=51.
- **Production (W=5, K=2)**: fires **36.0%**, accuracy **57.6%**. Under the always-up baseline of **66.0%** by 8.3 pp.
- **Winner (W=10, K=2)**: fires **51.9%**, accuracy **66.9%**. Above baseline. Softer score distribution (median |s| = 0.20 vs 0.40 in production) — more graded.
- **Binding knob is W, not K.** Holding K=2, accuracy climbs 57.6 → 63.2 → 66.9 → 64.7 across W=5/7/10/15.
- **K=1 is noise.** Fire rate drops with W but accuracy hovers at 56-61% — single-bar signals don't carry directional information.
- **High-conviction corner:** W=15, K=1 hits 82.9% accuracy but only 5.5% fire rate (44 sessions). Too rare to ship as primary, documented as a future overlay.

## Headline figure — W × K grid



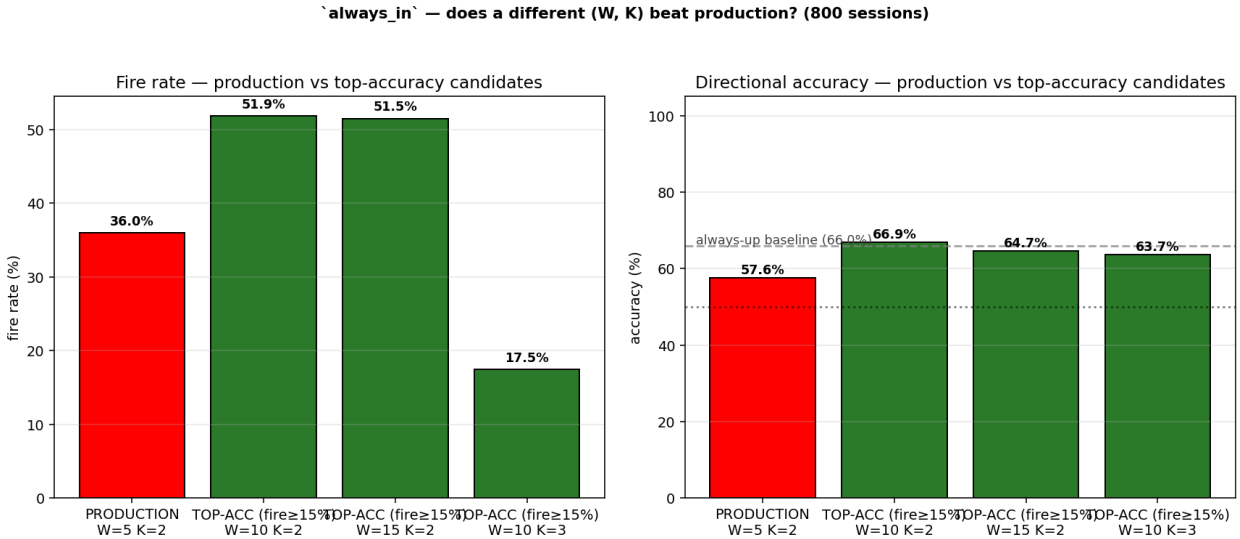

*Three heatmaps — fire rate, directional accuracy, median |score|. Rows = ALWAYS_IN_WINDOW, columns = DIRECTION_MIN_CONSEC. Red box = production cell. The middle panel (accuracy) is the story — production sits at 58%, W=10/K=2 at 67%.*

## Pareto view — fire rate vs accuracy




*Every (W, K) cell plotted. Dashed grey = always-up baseline; cells above add information, cells below subtract it. Production sits below baseline. W=10, K=2 sits above baseline with 52% fire rate.*

## Production vs top-accuracy candidates (fire ≥ 15%)




*Only cells with fire rate ≥ 15% are eligible so we're not cherry-picking rare high-accuracy cells. W=10, K=2 wins both axes.*

## What's next — needs your nod

- **PRIMARY (new):** change **ALWAYS_IN_WINDOW: 5** → **10** in aiedge/context/trend.py and the mirror **STRONG_TREND_WINDOW** in aiedge/signals/components.py. Re-baseline **TrendStateAlwaysInContributor** replay tests. Re-run the incr 19 degeneracy audit to confirm always_in exits the below-baseline zone.
- **STILL PENDING (incr 18):** lower MAJORITY_TREND_BAR_FLOOR to 0.25. Independently motivated — both sub-baseline contributors need recalibration.
- **STILL PENDING (incr 19):** docstring patch on compute_trend_state flagging the silent-zero htf_alignment trap for offline studies.
- **NEXT STUDY:** same-style sweep on **day_type** — incr 19 flagged it as 19.6% fire on this same sample.
- **NEXT STUDY (bigger):** rebalanced 400-up / 400-down sample across a longer window so the 'beats baseline' claim isn't dependent on this 9-day up-biased slice.
- **Caveat:** 'directional accuracy' here is same-session open→close labelling fidelity, not forward WR. Pattern Lab WR-driven weighting (still blocked on DB backfill) remains the right path to update equal weighting in production.

Findings site: https://github.com/zerosumsystems-ui/iPhone-/tree/main/trend-classification